

Integrated Biodiversity Monitoring on a Seaweed Farm

A general, affordable, multi-method approach that folds biodiversity monitoring into routine seaweed-farm operations — covering settled epibionts, mobile fish and invertebrates, and cetaceans — with environmental context.

Protocol ID: berger_2024

Sector: Seaweed · Usability Tier 4 (Practitioner)

MEL indicators: H&B 1.1.1 (Epibiota) · H&B 1.2.1 (Mobile fauna) · H&B 2.1.1 (Community composition)

Source paper: Integrating environmental and ecological monitoring with seaweed farming

Berger, C., et al. (2024). bioRxiv (preprint). DOI: <https://doi.org/10.1101/2024.02.15.580450>

Introduction

This protocol folds biodiversity monitoring into routine seaweed-farm work using a set of affordable, complementary methods that together span the range of organisms a farm affects — from stationary epibionts on the lines and crop, through mobile fish and invertebrates around the farm, to cetaceans passing through. No single method captures all of these, so three are combined, each suited to a different size/mobility band, alongside basic environmental measurements that provide context for the biodiversity observations.

The methods are chosen to be low-cost and integrable into normal farm operations rather than to require specialist survey campaigns. Run whichever combination fits your goals and gear; the sections below are modular. All equipment is described by role — substitute your own.

Measured parameters

- **Epibiont & seaweed community** — taxa and % area cover on header/cultivation lines and on the crop, plus biofouling cover on the crop blades. (*H&B 1.1.1, H&B 2.1.1*)
- **Mobile fauna** — fish and large-invertebrate species identity, relative abundance (Nmax = peak simultaneous count in a video sample), and length. (*H&B 1.2.1*)
- **Cetaceans** — detection-positive minutes (DPM) per hour and per day, per species, from a static acoustic detector; residency time. (*H&B 1.2.1*)

Context parameters (recorded, but not mapped to an MEL indicator): water temperature, seabed irradiance / light, current speed, pH, salinity and nutrients, plus crop growth/morphology and seabed shading. These characterise the site and help interpret the biodiversity data; they are not themselves framework indicators.

Requirements

Personnel: 1–2 people with a small vessel; the methods are designed for time-constrained operators. *[estimate]*

- Direct observations & BRUV: a vessel visit each round *[e.g. monthly]*; BRUV soaks are short *[e.g. 1 h]* each.
- Acoustic detector: deployed continuously for weeks-to-months per period; serviced periodically.
- Environmental loggers: deployed continuously *[e.g. logging every 20 min]*; swapped/serviced periodically; monthly water samples.

Sampling frequency & timing

Framework guidance. The MEL framework sets a recommended sampling frequency for each indicator, on the principle that sampling should be spaced “at intervals sufficient to build an accurate depiction of the process over time.” Its values run Once · Seasonal · Biannual · Annual · Ongoing.

What the source did. Ran the visual and video methods roughly monthly across more than a year, with the acoustic detector deployed for several multi-week periods spanning the seasons; environmental loggers ran continuously with monthly water samples.

Recommended for sound science. Run each method at least monthly — or at minimum once per season — across a full year, because epibiont and faunal communities turn over seasonally and a few visits cannot represent the assemblage. Sustain it over multiple years [*set to your baseline / impact goal*]: a first year sets the baseline, and only repeated years separate a farming-related change from natural interannual variability.

Materials

Direct observations (epibionts & crop)

- A vessel and a cutting tool to lift and sample line sections; sample containers; a way to estimate % area cover and identify taxa [*use preferred quadrat / visual-estimation approach*].

Baited Remote Underwater Video (BRUV)

- A baited underwater video rig [*use preferred camera on a weighted frame*]; a bait bag with oily/crustacean bait [*use locally available bait, ~500 g*]; a combination of floating + leaded rope to isolate the rig from surface motion; an anchor.

Passive Acoustic Monitoring (PAM)

- A static cetacean click-detector [*use preferred acoustic logger, e.g. a POD-type detector*] and its analysis software; a mooring using floating + leaded rope and an intermediate anchor to hold it steady in current.

Environmental context

- Temperature / current and light loggers [*use preferred loggers*], with a control light logger away from the farm for shading; a water-sampling bottle and a pH / salinity meter [*use preferred gear*].

Methods

A. Direct observations of lines and crop

1. Along fixed line-section lengths [*e.g. 0.5 m*], lift the header/cultivation line onto the vessel and cut the seaweed off close to the line [*~2 cm*]. Sample at more than one location per line [*e.g. two ends, ~90 m apart*].
2. For each section, subsample individuals at random [*e.g. 10*] and estimate the % area cover and taxa of epibionts on the line and on the crop.
3. Separately, record biofouling on the crop blades: the fouling taxa (e.g. hydroids, filamentous algae, bryozoans, bleaching) and their % blade cover, tracked through the season.

B. Baited Remote Underwater Video (BRUV)

1. Bait the rig and deploy it on the seabed near the farm in daylight, away from cliff/edge effects; isolate it from surface motion with the floating + leaded rope. Soak for a fixed duration [*e.g. 1 h*].
2. Classify footage quality first; analyse only footage where the bait bag is clearly visible and organisms are identifiable at least to family. [*set your visibility / ID threshold*]

3. For each species, score Nmax (the peak number seen simultaneously, which avoids double-counting) and measure length where possible.

C. Passive Acoustic Monitoring (PAM)

1. Deploy the static click-detector on the mooring near the farm; hold it steady in current (a vertical, well-anchored arrangement improves detection). Log continuously over the deployment period.
2. Extract detections with the detector's train-detection software; keep only detections classified as high-confidence cetacean origin. Export DPM per hour and per day for each species, and compute residency time per visit.

Environmental context

1. Deploy the temperature/current and light loggers at the site [*e.g. every 20 min*]; place a control light logger away from the farm to measure seabed shading. Collect water samples [*e.g. monthly*] and measure pH and salinity soon after collection.

Analysis

Handled in the paired stats workbook, per method. Simple summaries — % cover by taxon, mean Nmax and species richness / Shannon diversity, DPM aggregates — are done in the workbook. Differences in community *composition* between sites or over time are multivariate (NMDS / PERMANOVA) and belong in R; the workbook's *Stats directions* sheet gives a recipe.

Expect these failure modes (and manage them)

- **BRUV** — turbidity cuts detection (classify footage, use only clear frames); bait biases toward generalist carnivores and under-detects cryptic/juvenile flatfish; Nmax can be inflated by species that linger at the bait.
- **PAM** — gives relative presence, not abundance; detection depends on deployment geometry; boat traffic can confound diurnal patterns; a single detector covers a small area (patterns can differ within ~1 km), so deploy several for larger sites.
- **Direct observations** — sampling is destructive; seeding and handling introduce variability; epibiont and biofouling cover are strongly seasonal, so keep the schedule consistent.
- **Environmental** — swap/clean loggers on schedule; nearby sites can read near-identical, so confirm before treating them as independent.

How this serves the MEL framework

- **H&B 1.1.1 — Epibiota:** direct — epibiont cover/taxa on lines and crop, plus crop biofouling.
- **H&B 2.1.1 — Community composition:** direct — seaweed species composition on the header lines over the season.
- **H&B 1.2.1 — Mobile fauna:** direct — BRUV for fish and large invertebrates; PAM for cetaceans.

Context only (not mapped): temperature, irradiance, current, pH, salinity, nutrients, crop growth/morphology and seabed shading are recorded to interpret the biodiversity data, but do not each map to an MEL indicator.

Data handling

- Label video/audio files and samples by method, site and date; keep the raw footage/recordings and the environmental logs as the raw record.

Normalized by AquaPIP from Berger et al. (2024), a preprint (not yet peer-reviewed). Section lengths, subsample counts, BRUV soak duration and bait, detector type, and logger intervals are site/gear choices, shown in [brackets]. The detailed sub-methods are defined by cited sources — the BRUV / Nmax technique (Unsworth et al. 2014) and the acoustic detector's own documentation — rather than this article. All three database indicators map cleanly here; no indicator was pruned.

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